Science

RESEARCH ARTICLES

Cite as: H. Zhang *et al.*, *Science*
10.1126/science.adr8470 (2025).

Deep generative models design mRNA sequences with enhanced translational capacity and stability

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Despite the success of mRNA COVID-19 vaccines, extending this modality to more diseases necessitates substantial enhancements. We present GEMORNA, a generative RNA model that utilizes Transformer architectures tailored for mRNA coding sequences (CDSs) and untranslated regions (UTRs), to design novel mRNAs with enhanced expression and stability. GEMORNA-designed full-length mRNAs exhibited up to a 41-fold increase in firefly luciferase expression compared to an optimized benchmark in vitro. GEMORNA-generated therapeutic mRNAs achieved up to a 15-fold enhancement in human erythropoietin (EPO) expression and substantially elicited antibody titers of COVID vaccine in mice. Additionally, GEMORNA's versatility extends to circular RNA, substantially enhancing circular EPO expression and boosting anti-tumor cytotoxicity in CAR-T cells. These advancements highlight deep generative AI's vast potential for mRNA therapeutics.

Messenger RNA (mRNA) vaccines have proven effective in preventing COVID-19 (*1*, *2*). There are numerous efforts to extend mRNA therapeutics to other indications (*3*, *4*), but stronger and longer-lasting protein expression is necessary for these applications to be successful (*5*). Designing optimal mRNA sequences, including the coding sequences (CDSs) and untranslated regions (UTRs), is of paramount importance to realizing the broad potential of mRNA therapeutics via enhancing their translational capacity (*6*), but remains challenging due to the extensive potential mRNA sequence space (Fig. 1A). Within this extremely large space, mRNAs exhibit wide differences in expression-related properties. Moreover, the presence of multiple optimization metrics with complex interdependencies, along with hidden objectives linked to implicit cellular mechanisms, adds to the complexity of this problem.

In past decades, numerous efforts have sought to address the challenges of mRNA sequence optimization. Previous studies have demonstrated that optimizing nucleotide usage, such as GC content and U percentage, or codon usage, for example via the codon adaptation index (CAI), can enhance mRNA translation efficiency (*7*–*9*). However, these methods oversimplify the problem by prioritizing a single optimization objective, focusing primarily on local optimization of individual nucleotide and codons without considering sequence and structural context. To incorporate richer contextual information, deep learning-based models have been proposed for CDS optimization (*10*, *11*), both utilizing Long Short-Term Memory (LSTM) networks. However, these models face limitations due to LSTM's insufficient capacity in processing long gene sequence and also inefficiencies arising from non-parallelized training, which constrain the training data size and hindering model generalizability (*12*). A recent study incorporated structural features into the optimization objective, enabled global optimization via dynamic programming (*13*), and demonstrated a 128-fold increase in antibody response to COVID-19 mRNA vaccines compared to a codon-optimized baseline. However, this algorithm has not been adapted to mRNA sequences with chemical modifications (*13*, *14*), resulting in mismatches and reduced effectiveness in designing sequences for therapeutic mRNAs (*6*, *15*).

It has been challenging to design 5′ UTRs de novo since the mechanisms governing 5′ UTR regulation of mRNA translation are not fully understood. Recently, Zeng *et al.* developed a 5′ UTR design approach based on minimizing 5′ UTR secondary structure that demonstrated enhanced translational efficiency in cell-based assays (*16*). However, other factors that may influence initiation efficiency, such as mean ribosome load (MRL) and UTR-CDS interaction, were not considered (*17*). Several groups have described machine learning-based methods for 5′ UTR design (*18*, *19*), which initially trained predictive models on labeled data and then incorporated them into genetic algorithms for 5′ UTR sequence evolution. These methods depend heavily on the reliability of the predictive models, and the evolutionary algorithms can be hindered by local optima.

We developed generative models to design mRNA sequences with improved translational capacity. Drawing parallels between human language and genetic language, we adapted the concept and architecture of generative models from text generation to mRNA design. Specifically, designing a CDS for a given protein is analogous to generating a translated sentence from its source language, and designing an UTR is akin to freely composing a poem with accurate grammar and semantics (fig. S1). The generative model GEMORNA, which stands for Generative models for RNA, is the core of our mRNA design process. By leveraging Transformer-based RNA language models specifically tailored for CDS and UTR, GEMORNA extracts key optimization features from gene sequences, ensuring that the generated mRNA sequences fall within a high-potency space (Fig. 1B).

The effectiveness of GEMORNA-generated sequences was confirmed through extensive in vitro and in vivo experiments. We performed cell-based experiments to compare GEMORNA-generated CDSs and UTRs to a diverse set of widely used controls, including natural, algorithm-optimized, and commercial sequences expressing both reporter and therapeutic proteins. Based on these AI-designed CDSs and UTRs, our full-length mRNAs on various targets achieved substantial improvements in translational capability in vitro, as well as enhanced expression and immunogenicity in vivo. Moreover, GEMORNA's versatility extends to circular RNA (circRNA) design. For example, we used GEMORNA to create CD19 chimeric antigen receptor (CAR)-expressing circRNAs and showed that non-virally transduced T cells exhibited substantially stronger tumor-killing efficacy compared to a patented circRNA benchmark (*20*). Taken together, our generative AI models enable the de novo generation of linear and circular RNAs with enhanced translational capacity, and have the potential to improve the potency of mRNA drugs across a variety of therapeutic areas.

GEMORNA-CDS: deep generative model for CDS design

We first sought to develop generative models for CDS design, as CDSs are the core of mRNA therapeutics and vaccines. Drawing upon the analogy between natural language translation and CDS design, we inferred that classical sequence-to-sequence (Seq2Seq) models (*21*), originally designed for human language translation tasks, could be adapted for transforming genetic sequences. Specifically, we treated protein sequences as the source language and mRNA CDSs as the target language, utilizing a Transformer architecture (*22*) with an encoder module to process protein information and a decoder module to generate CDS sequences. During training, natural CDS sequences were tokenized and fed into the decoder with a causal mask for teacher-forced learning, while the decoder outputted codon probability distributions in parallel (Fig. 2A). Cross-entropy loss was computed between input and output distributions and backpropagated to update model parameters. At inference, teacher-forced decoding was replaced with autoregressive decoding, employing strategies such as sampling, greedy search or beam search to generate diverse CDS sequences (Fig. 2B). We observed that some key features of GEMORNA-generated CDSs, such as CAI (*9*) increased progressively from unbiased sampling to biased sampling and further with greedy and beam search decoding. This indicates that, despite being trained on natural CDSs, GEMORNA can generate sequences that deviate from natural ones (fig. S2).

GEMORNA CDSs demonstrate advantageous properties across multiple optimization-related features

Recent studies suggested that multiple features, including codon usage, GC content, and mRNA structure, should be considered for CDS optimization (*23*, *24*). Following this, we analyzed over 3,000 GEMORNA-generated CDSs, each corresponding to a mammalian protein produced via greedy decoding, and compared them to their natural, randomly-generated and CAI-optimized counterparts across eight optimization-related properties (Fig. 2C).

Regarding codon and nucleotide usage, GEMORNA CDSs exhibited higher CAI and GC content while maintaining lower rare codon rate and U percentage. These characteristics aligned with principles for designing therapeutics mRNAs that potentially promote translation efficiency and minimize unwanted innate immunogenicity (*8*, *25*, *26*). Additionally, we found that GEMORNA-designed sequences consistently disfavor unstable codons, and lead to a higher correlation with codon stability coefficient (CSC) (*27*) (fig. S3). GEMORNA was trained in an unsupervised manner without pre-screening sequences based on CAI, GC content, or U percentage thresholds, indicating that it autonomously learned codon and nucleotide usage through data-driven modeling.

In terms of mRNA structure, GEMORNA-generated CDSs contained more secondary structures, reflected by lower minimum free energy (MFE), compared to natural sequences and CAI-based designs, yet exhibited fewer secondary structures than MFE-optimized sequences from LinearDesign (Fig. 2D and fig. S4). Notably, the length-normalized MFE of most GEMORNA CDSs fell within the range observed in endogenous mammalian CDSs.

Although two individual codons may each have moderate frequencies, their joint occurrence could be infrequent due to inhibitory interactions, resulting in "unwanted codon pairs" (*28*, *29*). Notably, natural sequences tend to have low frequencies of unwanted codon pairs, while CAI-optimized sequences contain more unwanted codon pairs (Fig. 2C). In contrast, our AI-generated CDSs exhibited a marked reduction in the incidence of unwanted codon pairs. Furthermore, we observed that AI-designed CDSs contained fewer slippery sites (*30*, *31*), reducing the +1 ribosomal frameshifting frequency

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(Fig. 2C). These findings confirm that the GEMORNA model has the capability to learn CDS patterns within a contextual framework.

Beyond these features, we also examined the naturalness score, which captures more complex and implicit sequence patterns within mammalian CDSs. This score, adapted from the concept of perplexity in natural language processing, has been previously applied to gene sequences (*32*, *33*). Specifically, naturalness quantifies how well a CDS sequence aligns with the genetic patterns of the mammalian genome, as inferred by a language model. We observed that GEMORNA-generated CDSs exhibited a higher naturalness score distribution compared to other control groups (Fig. 2C), suggesting that these sequences were more aligned with the mammalian genomes in the training data.

The complexity of CDS optimization arises in part from the intricate interdependencies among optimization objectives. To analyze these interdependencies, we visualized them in two-dimensional plots (Fig. 2D). The CDSs from different sources formed district clusters in most 2D plots, suggesting varying patterns in sequential and structural features among random, natural and synthetic CDSs. The natural CDS clusters consistently shifted to the upper-right direction of the random CDS clusters, aligning with the trajectory of joint optimization of two objectives. 有猫腻? This likely reflects a directional adaptation of mRNA sequences in mammals. Notably, CAI-optimized and GEMORNA-generated CDSs extended along these adaptive patterns, while GEMORNA CDSs were positioned farther from natural groups than CAI-optimized sequences. 小结论 This suggests that GEMORNA assigned higher probabilities to mammalian-favored codons. In contrast, CDSs generated by the LinearDesign algorithm (*13*) displayed distinct characteristics in most plots, highlighting how MFE-driven optimization diverged from broader mRNA patterns observed in nature.

Naturalness and other metrics correlate with experimentally validated expression and stability data

To establish a link between in silico metrics and in vitro CDS performance, we analyzed their correlations with expression and stability data from multiple sources. Specifically, we incorporated two independent datasets: (1) the Bicknell *et al.* dataset, which assessed m1Ψ-modified exogenous mRNAs in human cells as well as mice (*23*); (2) the Leppek *et al.* dataset, which examined unmodified exogenous mRNAs in human cells (*15*). We compared the correlations of various metrics with experimental data, including those shown in Fig. 2, C and D, as well as a joint metric (MFE-CAI) combining structure and codon optimality, following previous studies (*13*, *23*, *34*).

For the Bicknell dataset, most metrics, except for CAI, demonstrated strong correlations with expression and stability (Fig. 2E). Notably, naturalness score exhibited the strongest correlation, achieving *r* values of 0.92 and 0.96 for in vitro half-life and in vivo expression, respectively. In contrast, for the Leppek dataset, overall correlations were lower. The nucleotide usage metrics showed the highest correlations, with U percentage correlating most at the 6-hour time point and GC content at 24 hours. This suggests that GC richness and U depletion are key factors in optimizing unmodified mRNAs, potentially due to the innate immune response triggered by exogenous uridines. 可能由于外源性尿苷触发的固有免疫反应

We also incorporated an in-house dataset evaluating m1Ψ-modified mRNAs encoding destabilized firefly luciferase (Fluc2P) in HEK293T cells. This dataset further confirmed that U percentage and naturalness were the most correlated factors with in vitro expression at 24 and 48 hours. On average, naturalness demonstrated the strongest correlation across different data sources, slightly surpassing nucleotide usage metrics (Fig. 2E). This suggests that GEMORNA's ability to achieve zero-shot generation of mRNAs with high naturalness, along with other beneficial metrics, may contribute its success in generating mRNAs with strong performance.

GEMORNA-generated CDS demonstrated stronger and long-lasting expression

Next, we validated these results with Fluc2P, a luciferase variant with degradation-promoting mutations (*35*). We first generated four CDS sequences, labeled from GMR-FL-CDS1 to GMR-FL-CDS4, using the GEMORNA-CDS model, employing both greedy and sampling decoding. These four CDSs were then compared against five controls: codon-optimized CDSs from online tools of GenScript (CAI-G) and IDT (CAI-I), a CAI- and structure-jointly optimized CDS from LinearDesign (LinearDesign) (*13*), a deep-learning-based CDS from a biLSTM-CRF model (biLSTM-CRF) (*10*), and a CDS from a commercially available vector (pGL4.11). Our AI-designed CDSs exhibited up to a 20-fold improvement over biLSTM-CRF, and a 4.8-fold over pGL4.11 48 hours after transfection (Fig. 2F). Given the rapid degradation of destabilized luciferase post-production, the Fluc activity ratio at 48 hours relative to 24 hours was used as a metric to evaluate the in-cell stability of mRNA constructs. GEMORNA-designed sequences exhibited higher Fluc activity ratio, suggesting they have greater in-cell stability (Fig. 2G). These experiments performed in HepG2 cells showed consistent results with those generated in HEK293T cells (fig. S5).

GEMORNA-UTR: deep generative model for UTR design

In addition to mRNA CDSs themselves, the UTR sequences that surround the CDSs play major roles in the performance of mRNA therapeutics and vaccines by controlling expression levels and mRNA stability (*36*, *37*). We built the

GEMORNA-UTR model to generate UTRs by stacking Transformer decoders rather than using an encoder-decoder architecture as with the GEMORNA-CDS model (Fig. 3, A and B). A notable advantage of this decoder-only architecture is its capability for auto-regressive generation of nucleotides with no reliance on source information, which is more suitable for de novo UTR generation, as UTRs are not constrained by protein-coding requirements like CDSs (Fig. 1A).

The GEMORNA-UTR model was first pre-trained on natural UTRs, which allows it to mimic and learn implicit rules from natural genetic sequences. Then we fine-tuned GEMORNA-UTR with selected natural UTRs that tend to possess higher translation efficiency or stability. This makes a difference in the training methodology between the GEMORNA-CDS and GEMORNA-UTR models. Specifically, GEMORNA-CDS was trained without a fine-tuning process; instead, by applying more biased sampling strategies at inference time, such as greedy decoding, it can generate CDSs with the desired properties and superior performance (Fig. 2, C to G). In contrast, GEMORNA-UTR undergoes fine-tuning to better learn sequence patterns, context-dependent motifs and functional properties, as these features are highly diverse or sparsely distributed across natural UTR datasets used for pre-training.

To do fine-tuning, we used 5′ UTRs with high mean ribosome loads (MRLs), since MRL is considered to be correlated with translation efficiency (*18*). We built an MRL-predictive model with high accuracy and identified 5′ UTRs to be used for fine-tuning based on this model (Fig. 3C and fig. S6). Similarly, we trained a predictive model that estimates the stability contributed by 3′ UTRs based on previously published experimental data (fig. S7) (*38*), and identified 3′ UTRs for fine-tuning.

GEMORNA-UTR generated novel UTRs with higher MRL, MFE and enhanced protein expression

We first assessed the similarities between GEMORNA-generated UTRs and natural UTRs using BLAST. Maximum Identity Score (MIS) distributions suggest that the GEMORNA-UTR AI model, with or without fine-tuning, is able to generate novel UTRs that are dissimilar to natural UTRs (Fig. 3D and fig. S7). Next, we evaluated the MRL and MFE of GEMORNA 5′ UTRs. Prior to fine-tuning, the predicted MRLs and MFEs of GEMORNA 5′ UTRs showed considerable concordance with natural ones, which suggests that GEMORNA-UTR can generate 5′ UTRs that mimic the properties of natural ones. Post-fine-tuning, the MRL and MFE distributions of GEMORNA-generated 5′ UTRs shifted rightwards to higher values (Fig. 3D), which can potentially lead to high efficiencies in translation initiation.

We experimentally compared GEMORNA-generated 5′UTRs against a comprehensive benchmark of diverse 5′ UTRs, covering natural UTRs and natural derivatives, structure-based rational designs from previous studies, genetic algorithm-generated designs with machine learning predictors, and UTRs used in commercial mRNA drugs (*15*, *16*, *39*, *40*). To isolate the effects of the 5′ UTR, we used Fluc2P as the target protein and kept the CDS and 3′ UTR constant across all constructs. Among the comparison benchmarks, the 5′ UTRs used in BNT162b2 COVID mRNA vaccines showed the highest Fluc activity in HEK293T (Fig. 3E). We assessed 12 GEMORNA-designed UTRs for their ability to enhance protein expression and confirmed that five GEMORNA 5′ UTRs exhibited similar or higher Fluc activities compared with BNT162b2 5′ UTR (Fig. 3F). The performance of 5′ UTRs in HepG2 cell line showed a strong correlation with their performance in HEK293T, with an r^2 value of 0.92 (Fig. 3G).

Similarly, we designed ten 3′ UTRs using GEMORNA-UTR and benchmarked them against natural, algorithm-designed, and commercial mRNA drug 3′ UTRs (fig. S8). Many sequences, including those from GEMORNA and the benchmarks, exhibited comparable expression levels, with less variation than 5′ UTR benchmarking, likely because 3′ UTRs play a secondary role in translation efficiency. Notably, eight of the ten GEMORNA 3′ UTRs matched or outperformed BNT162b2's 3′ UTR, offering viable de novo alternatives.

5′ UTRs regulate expression levels in combination with 3′ UTRs

The influence of 5′ UTRs on expression levels varies depending on the 3′ UTR with which it is paired (*41*). To investigate this relationship, we examined the effects of various 5′ UTR and 3′ UTR pairs with a fixed CDS, using two different CDSs encoding Fluc2P reporter and CD19 CAR (Fig. 3H). Specifically, we paired eight 5′ UTRs with eleven different 3′ UTRs, forming eight groups where each group shared the same 5′ UTR. Overall, more than 80% UTR pairs demonstrated improved expression compared to the BNT162b2 benchmark (Fig. 3I). For Fluc2P, fig. 3J demonstrated that the average Fluc activity across groups closely aligned with the 5′ UTR performance shown in Fig. 3F. Within each group, different 3′ UTR introduced variances in expression levels, with greater variability observed at higher expression levels. This is likely due to the 3′ UTR's role in stability and localization, as well as interactions with the 5′ UTR (*42*). Additionally, utilizing these UTR pairs led to an up-to 7-fold increase in Fluc activity compared to the benchmark BNT162b2 UTRs. For CD19 CAR, the influence of UTR pairs were also evident within each group (Fig. 3K).

To further examine the contributions of 5′ and 3′ UTRs to expression levels, we plotted the average fold-change for each group, where the 5′ UTR is constant while the 3′ UTR varies (i.e., the mean expression levels represented by short lines in Fig. 3, J and K), against the corresponding values for the same 5′ UTR when paired with the BNT162b2 3′ UTR. The results in Fig. 3L revealed a correlation between the group average fold-change observed with varied 3′ UTRs and that observed with a fixed BNT162b2 3′ UTR, indicating that the 5′ UTR primarily drives expression regulation. On the other hand, UTR pairs did not perform consistently across both protein targets, with a low correlation (r^2 = 0.096) (Fig. 3M). This indicates that UTR regulation is highly target-dependent, making it unlikely that a universally optimal UTR pair exists. Consequently, this highlights the need for generative models capable of producing diverse 5′ and 3′ UTRs, combined with accurate and high-throughput experimentation to validate optimal designs for specific applications.

Full-length mRNA design with GEMORNA

In previous sections, we showed that both GEMORNA-generated CDSs and UTRs achieve higher in vitro translational capacity compared to a variety of baseline sequences. Next, we investigated whether full-length mRNA designs, derived from GEMORNA elements, could further enhance protein expression and lead to better immunogenicity in mRNA vaccines. To conduct a comprehensive analysis, we examined these questions with four distinct protein targets: Fluc2P reporter, COVID-19 spike protein as an antigen for mRNA vaccine, NanoLuc luciferase (NanoLuc) reporter, and human erythropoietin as a representative for gene replacement therapy (Fig. 4A). Notably, for Fluc2P and COVID-19 targets, we utilized a two-step design process: first, we designed and validated an optimal CDS, then integrated it with various UTR pairs for a full-length mRNA evaluation. In contrast, for NanoLuc and EPO, we directly designed and validated full-length mRNA sequences without a separate CDS selection step.

GEMORNA full-length mRNA reporters achieved enhanced protein expression and durable expression

For the Fluc2P reporter CDS sequence, we used the GMR-FL2 across all full-length mRNA designs since it had the strongest translational capacity out of four GEMORNA CDSs we tested (Fig. 2G). For the UTRs, we randomly selected three 5′ UTRs and three 3′ UTRs from validated GEMORNA UTRs to create five random UTR pairs for full-length mRNAs that were then compared against two benchmark mRNAs: the first consisting of an IDT-designed CDS with natural alpha globin (AG) UTRs (Benchmark-FL1), and the second consisting of the pGL4.11 CDS with BNT162b2 UTRs (Benchmark-FL2). We observed that all GEMORNA-designed sequences achieved significantly higher Fluc activity than the benchmarks, with a more pronounced enhancement at 48 hours compared to 24 hours (Fig. 4B). Specifically, GMR-FL-F5 demonstrated a 41-fold increase in expression compared to Benchmark-FL1 and an 8.2-fold increase relative to Benchmark-FL2. Protein expression improvements were also observed in HepG2 cells. Remarkably, GMR-FL-F5 achieved a 15.9-fold increase relative to Benchmark-FL2 (Fig. 4C).

GEMORNA-designed mRNA vaccine demonstrated higher antigen expression and immunogenicity

We further validated the GEMORNA model on synthetic CDSs for COVID-19 mRNA vaccines. We observed that GEMORNA-generated CDSs outperformed both LinearDesign Seq-C (*13*) and the commercial BNT162b2 (Fig. 4D). The benefits were more pronounced when the top-performing GEMORNA CDS was combined with GEMORNA UTR pairs in full-length designs (Fig. 4E). Next, we tested the immunogenicity of our full-length designs against the benchmarks. Lipid nanoparticle (LNP)-encapsulated mRNAs were injected into five BALB/c mice for each sequence (Fig. 4F). GEMORNA-derived full-length mRNA induced higher antibody titers than both the BNT162b2 and the LinearDesign mRNAs at multiple time points post-immunization (Fig. 4, G and H and fig. S9). These results confirm the generalizability of GEMORNA model to different proteins, including therapeutically relevant ones.

GEMORNA can directly design full-length mRNAs with strong expression

Directly designing and experimentally validating superior full-length mRNAs without first identifying the best CDS would accelerate mRNA development process. To explore this approach, we combined five GEMORNA CDSs with three GEMORNA UTR pairs, and randomly selected seven full-length designs. This strategy was applied and evaluated with two targets: NanoLuc and EPO.

For the NanoLuc luciferase reporter, GEMORNA-designed mRNAs exhibited higher expression levels than a strong benchmark sequence that utilizes BNT162b2 UTRs and a CDS from the literature (*39*), with improvements observed from 24 to 72 hours (fig. S10). For EPO, seven GEMORNA-created mRNAs were compared to a benchmark sequence that consisted of codon-optimized CDS (*39*) and BNT162b2 UTRs; six of the GEMORNA designs achieved enhanced EPO activities (Fig. 4I). Notably, our design, GMR-EPO-F6, demonstrated expression that lasted longer than the benchmark (Fig. 4J). GEMORNA designs also showed higher expression levels in HepG2 cells (Fig. 4K). We subsequently selected three EPO designs with the highest in vitro expression for in vivo validation. GEMORNA sequences demonstrated longer durability and stronger expression levels than the benchmark in mice (Fig. 4L). Notably, GMR-EPO-F7 achieved 15-fold increase in expression compared to the benchmark at 24 hours.

Circular RNA design with GEMORNA

CircRNAs exhibit increased resistance to exonucleases compared to their linear RNA counterparts (*43*). Here we showed GEMORNA is extensible to circRNA design, generating circRNAs with enhanced durability, improved expression levels, and strong efficacy.

GEMORNA-generated circRNAs demonstrated high and long-lasting expression

Initial tests using Fluc2P luciferase assays revealed that GEMORNA-generated CDSs outperformed conventionally designed CDSs within circRNAs (fig. S11). In addition to CDS optimization, a previous study demonstrated enhanced circRNA expression with optimized topology, screened UTRs and engineered IRES (*39*). We hypothesized that the performance of circRNA could be further enhanced by AI-generated CDSs and mined IRESs, even with a simplified topology (Fig. 5, A and B). To verify this, we conducted additional experiments with NanoLuc luciferase assays, using a systematically optimized circRNA (*39*) as the benchmark (labeled as Chen *et al.*)., while our designs maintained a simplified topology. All our designs exhibited substantially higher accumulated NanoLuc luciferase expression and strong expression durability (Fig. 5, C to E). One design, GMR-NL3, maintained a high NanoLuc luciferase activity over 144 hours, achieving a 3-fold increase in NanoLuc luciferase activity at 72 hours compared to 24 hours. In contrast, the NanoLuc luciferase activity of the benchmark continuously declined after 48 hours. We observed similar results in HepG2 cells (fig. S12).

We further evaluated circRNAs encoding EPO. Similarly, the benchmark sequence was also from Chen *et al.* (*39*), constructed with optimized IRES, UTRs and CDS. Figure 5F shows EPO expression levels from 24 hours to 144 hours in HEK293T cells and demonstrates that the best GEMORNA design exhibited a 13.8-fold increase in accumulated expression compared to the benchmark. Additionally, GEMORNA sequences showed superior ratios of expression levels at 144 hours compared to 24 hours, with the highest ratio being 46.5%, in contrast to the benchmark's 2.5%, thus achieving greater expression longevity (Fig. 5G). Similar results were observed in HepG2 cells (fig. S13).

Subsequently, we conducted in vivo evaluation on three GEMORNA circRNAs that exhibited strong translation capacity in vitro. All three GEMORNA circRNAs demonstrated significantly higher in vivo EPO expression levels at 24 hours compared to the benchmark, with the most effective circRNA showing a 121-fold increase (Fig. 5H). Additionally, our AI-designed sequences maintained long-lasting and higher expression levels from 6 hours to 168 hours in vivo.

GEMORNA circRNAs achieved enhanced efficacy for non-viral CD19 CAR-T therapy

Given the success of mRNA vaccines, interest in leveraging mRNA technology for in vivo CD19 CAR-T therapy has been growing,and it is currently under evaluation in clinical trials (*44*, *45*). We conducted non-viral CD19 CAR T experiments in vitro to assess the durability and the therapeutic potential of GEMORNA-enabled circRNAs for CAR-T therapy (Fig. 5I).

We first constructed two baseline constructs, one denoted as “Benchmark-1” where the coding region was designed using a conventional CAI-based approach, and the other denoted as “Benchmark-2”, a patented circRNA (*20*). The baselines and GEMORNA-designed circRNA were then delivered into human primary T cells. Our circRNA exhibited substantial increases in gene expression, surpassing “Benchmark-1” by 28-fold and “Benchmark-2” by 5.6-fold at 24 hours post-electroporation (Fig. 5J). Additionally, 50% of the cells with GEMORNA-designed circRNA were still CD19 CAR-positive 120 hours post-electroporation, compared to less than 72 hours for the benchmark circRNAs (Fig. 5K). We also confirmed GEMORNA circRNA showed robust improvements in gene expression level and tumor cell cytotoxicity over a linear mRNA control (fig. S14). We further investigated the cytotoxic efficacy of circRNA candidates in human primary T cells against NALM-6 (Fig. 5L). As shown in Fig. 5M and fig. S15, our highly expressive and long-lasting circRNA effectively eliminated the NALM-6 cells, while the codon-optimized benchmark showed virtually no cytotoxic effect. The benchmark sequence “Benchmark-2” was also less efficient than our design.

Our results indicate that generative AI-designed circRNAs effectively enhanced CD19 CAR-T cell cytotoxicity in vitro, and that this approach may have utility for in vivo CAR-T therapy.

Discussion

mRNA therapeutics have emerged as a promising next-generation modality, yet their broader application is hindered by low protein expression levels and short expression duration. To address these challenges, we propose GEMORNA, deep generative models designed to enhance mRNA translational capacity. Experimental validation confirms that GEMORNA-derived elements contribute to the development of superior mRNA drug molecules, offering distinct advantages over existing mRNA optimization approaches. First, GEMORNA-generated elements demonstrate enhanced translational capacity when combined with m1Ψ nucleotide, a critical modification to the success of marketed mRNA drugs (*14*, *46*). Second, GEMORNA is extensible to the design of circRNAs, leading to substantial improvements in long-lasting expression. Third, unlike prior RNA language models developed primarily for predictive tasks (*47–49*), GEMORNA is conceived as deep generative models specifically designed to directly generating high-quality mRNA elements. Finally, GEMORNA-derived sequences were benchmarked against commercial products and state-of-the-art designs, exhibiting substantial improvements in expression levels, durability and immunogenicity. Despite its strengths, GEMORNA could be enhanced with high-throughput expression and stability data for exogenous sequences. Additionally, GEMORNA's framework could support the development of specialized models tailored to tissue- or disease-specific applications, further broadening its utility. On the other hand, as a deep learning-based model, GEMORNA remains somewhat of a black box, making it difficult to interpret the implicitly learned features. Further validation will be valuable in gaining a deeper understanding as new insights into translation and degradation mechanisms emerge. In conclusion, our generative AI-based model constitutes a novel and efficacious strategy for mRNA sequence design, with the potential to notably advance the development of mRNA vaccines and therapeutics.

Materials and Methods

Data collection for GEMORNA models

The training dataset for the GEMORNA-CDS comprised over 1 million natural protein sequences and their corresponding CDSs from 115 mammalian species, sourced from Ensembl (*50*). All sequences were validated to ensure that each CDS was exactly three times the length of the corresponding protein sequence plus three, accounting for the stop codon. Sequences with mismatches in codon usage were further filtered to maintain consistency with the codon usage table. The pre-training dataset for GEMORNA-UTR was obtained from Refseq (*51*) and UTRdb (*52*). Mammalian sequences were selected, and filtering steps were applied to exclude long and redundant sequences, resulting in a final dataset of approximately 8 million 5′ UTRs and 2 million 3′ UTRs.

For fine-tuning, we applied filtering criteria based on predictions from the PRED-5UTR and PRED-3UTR models. Specifically, 5′ UTRs with high MRL predicted from PRED-5UTR and 3′ UTRs with high stability predicted from PRED-3UTR were selected and then randomly sampled for fine-tuning. This process results in fine-tuning datasets of approximately 800k samples for 5′ UTRs and 200k samples for 3′ UTRs, representing one-tenth of the size of their respective pre-training datasets.

GEMORNA model training

The default configuration of GEMORNA-CDS consists of a transformer architecture with 12 layers each for the encoder and decoder. Each layer includes 8 attention heads for self-attention and cross-attention. The hidden dimension is set to 128, and each layer is followed by residual connections, layer normalization, and fully connected feedforward networks.

GEMORNA-UTR employs 12 decoder layers with 12 self-attention heads per layer. Given that 3′ UTRs are generally longer and contain more complex functions and regulatory motifs compared to 5′ UTRs, we employed 144 hidden dimensions for 5′ UTR model, and a larger hidden dimension size of 288 for 3′ UTR model. As a result, the 3′ UTR model is approximately 4 times larger than the 5′ UTR model. The above models were trained on two NVIDIA L20 GPUs using mixed precision (FP16) to optimize memory and computation efficiency. The AdamW optimizer was employed with a learning rate of 5×10^{-4}. A linear learning rate scheduler was used with 1,000 warm up steps followed by decay over 400,000 total training steps. Each GPU processed a batch size of 32, with gradient accumulation over 4 steps to effectively increase the batch size. Dropout rate of 0.1 was applied to both attention and hidden layers for regularization.

GEMORNA-CDS model score and naturalness score

To evaluate the quality of the generated CDS sequences, we assign each sequence a log-likelihood score conditioned on the corresponding protein sequence, using this score as a zero-shot metric for CDS design. This model score M_s is computed similarly to the training objective, but in this case, it is applied to the generated CDS sequence $x = (x_1, x_2, ..., x_L)$ under the constrain of the protein sequence $y = (y_1, y_2, ..., y_L)$, which is given by

$$M_s(x, y) = \sum_{t=1}^{L} \log p_\theta(x_t \mid x_{<t}, y)$$

A higher model score indicates that the CDS sequence better fits the model's learned parameters and aligns more closely with the distribution of natural CDSs, under the constraint of the given protein sequence. Naturalness score N_s is the exponential form of the length-normalized model score:

$$N_s(x, y) = e^{\frac{1}{L} M_s(x,y)} = e^{\frac{1}{L} \sum_{t=1}^{L} \log p_\theta(x_t \mid x_{<t}, y)}$$

Maximum identity score calculation

The maximum identity score (MIS), defined as the longest aligned subsequence divided by the length of the query sequence, was employed as the measure of similarity. The reference library for BLAST search is composed of all UTRs used in the pre-training phase. The word size for alignment was set at 15 nucleotides. In BLAST search, the word size defines the shortest length of matching sub-sequences between the query and the database sequences.

Pred-5UTR: mean ribosome load prediction model for 5′ UTRs

We developed a model to predict the mean ribosome load (MRL) value for a given 5′ UTR sequence. This model incorporates two layers of unidirectional gated recurrent units (GRUs). The final states from these two layers are concatenated and followed by a fully connected layer. The input and output dimensions of the GRU model are 64 and 128, respectively. The training dataset (*18*) comprises 166,530 5′ UTR sequences with labeled MRL values. The sequence length varies from 50 to 100 nt, and all shorter sequences were padded on the right side. During training, the ratio of the training, evaluation, and testing sets is 0.7, 0.15, and 0.15, respectively. The learning rate is set to 0.0005, with a dropout rate of 0.1.

Pred-3UTR: stability prediction model for 3′ UTRs

We trained a TextCNN-based model (*53*) to predict the stability of 3′ UTR sequences. The dataset was sourced from Rabani *et al.*, comprising 90,000 3′ UTR sequences (*38*). Degradation rates were computed based on the experimental data for all sequences as the reference values. For the model, the embedding dimension is 256, and the kernel sizes are 2, 4, 6, 8, and 10, with a total of 200 kernels. No length limit was imposed. During training, the ratio of the training, evaluation, and testing sets was set to 0.7, 0.15, and 0.15, respectively. The learning rate was set to 0.0005, with a dropout rate of 0.1.

Plasmid construction

The plasmids used in this study were built using restriction enzyme cloning and Gibson assembly. Sequences of mRNA and circRNA constructs used in this study are detailed in tables S1 and S2.

mRNA synthesis

The IVT templates were amplified by PCR and purified with Zymoclean Gel DNA Recovery Kit (Zymo Research, D4008). The mRNAs were synthesized using T7 Co-transcription RNA Synthesis Kit (Synthgene, 10111-3011) with UTP fully substituted with N1-Methylpseudo UTP. Reaction mixtures were incubated at 37°C for 2 hours (1 hour for covid vaccine mRNAs), and the templates were removed with DNase I (NEB). The mRNAs were purified with the beads for the following studies.

CircRNA synthesis

The IVT templates were amplified via PCR and purified with Zymoclean Gel DNA Recovery Kit. The circRNAs were synthesized using HiScribe T7 High Yield RNA Synthesis Kit (NEB). Reaction mixtures were incubated at 37°C for 2 hours, and the templates were removed with DNase I. Additional GTP was added to a final concentration of 2 mM, and the reaction mixtures were incubated at 55°C for 15 min. The IVT products were purified with the beads and the residual linear RNA precursors were removed with RNase R (LGC Biosearch Technologies, RNR07250). The circRNAs were purified again with the beads for the following studies.

Cell culture and transfections

Mammalian cell culture experiments were conducted using HEK293T (ATCC CRL-3216), HepG2 (ATCC HB-8065), Jurkat (ATCC TIB-152), and NALM-6 expressing firefly luciferase- GFP (Shanghai Aowei Biotechnology, SAc0269LG) cells. HEK293T and HepG2 cells were cultured in DMEM (Gibco, 10569010) supplemented with 10% FBS (Gibco, A5669701) and 100 units/ml penicillin and 100 µg/ml streptomycin. Jurkat cells and NALM-6 expressing firefly luciferase-GFP (NALM-6) were cultured in RPMI 1640 (Gibco, 11875093) supplemented with 10% FBS (Gibco, A5669701) and 100 units/ml penicillin and 100 µg/ml streptomycin (Gibco, 15140122). HEK293T and HepG2 cells were seeded in 96-well plates one day before transfection, and the transfection was performed using Lipofectamine MessengerMax (Gibco, LMRNA015) according to the manufacturer's manual. Jurkat cells were electroporated using Gene Pulser Xcell Electroporation System (Bio-Rad) according to the manufacturers' manuals.

Firefly luciferase assays

To measure firefly luciferase activity, 100 µl of Bright-Glo Luciferase reagent (Promega, E2620) was added to each well, mixed, and incubated for 5 min. 100 µl of the mix was transferred to a flat-bottomed white opaque plate, and luminescence was measured using a microplate reader.

Nanoluc luciferase assay

At the indicated time points, 5-20 µl cell culture supernatants were transferred to a flat-bottomed white opaque and diluted with cell culture medium to a final volume of 100 µl. Equal volume of Nano-Glo Luciferase Assay Reagent (Promega, N1130) was added to each sample, and luminescence was measured using a microplate reader.

EPO ELISA assay

Erythropoietin levels were measured by ELISA (Abcam, ab274397) according to the manufacturer's instructions, except that cell culture supernatants collected at indicated time points post-transfection were used, and samples were diluted 1:200 prior to analysis.

T cell activation and expansion

Cryopreserved primary human peripheral blood mononuclear cells (PBMCs) were purchased from Shanghai Origin

Biotechnology Co., Ltd. (PB003F-C). PBMCs were thawed from cryogenic storage and were activated using coated anti-CD3 (BioLegend, 317301) and anti-CD28 (BioLegend, 302901) antibodies at 50 ng/ml. Activated T cells were expanded for 6 days in T Cell Expansion medium consisting of RPMI 1640 medium supplemented with 10% CTS Immune Cell Serum Replacement (Gibco, A2596101), 2 mM GlutaMAX supplement (Gibco, 35050061), 50 μM mercaptoethanol, 100 IU/ml IL-2 (PeproTech, 200-02), 100 units/ml penicillin, and 100 μg/ml streptomycin (Gibco, 15140122). Cell cultures were maintained at 37°C in a humidified 5% CO_2 atmosphere.

Generation of CD19-CAR T-cells by electroporation

In vitro expanded human primary T cells were electroporated with mRNAs or circRNAs using the Gene Pulser Xcell Eukaryotic System (Bio-Rad) according to the manufacturer's instructions. Prior to electroporation, expanded T cells were counted, assessed for viability using Trypan Blue, and washed twice with sterile phosphate-buffered saline. Cells were then resuspended at a density of 1×10^7 cells/ml in Gene Pulser Electroporation Buffer (Bio-Rad). Aliquots of the mixture were transferred into 1.5 ml microcentrifuge tubes (100 μl per condition). In vitro transcribed circRNAs was added to each mixture. The mixture was homogenized by pipetting and transferred into a sterile Gene Pulser/MicroPulser Electroporation Cuvette with 0.2 cm gap (Bio-Rad). Electroporation was performed using a square-wave pulse at 220 V for 2 ms. Immediately after pulsing, cells were transferred into a 24-well plate containing 900 μl prewarmed T Cell Expansion Medium.

Flow cytometry

Expanded T cells were electroporated with CD19 CAR mRNAs and stained for CD19 CARs with PE-labeled human CD19 (Acrobiosystems, CD9-HP2H3) at indicated time points. Cells wer co-stained with anti-human CD3 (BioLegend, 300412), anti-human CD4 (BioLegend, 344612), and anti-human CD8 (BioLegend, 344748), and analyzed on a BD FACSVerse flow cytometer. For flow cytometry analysis, lymphocytes were first gated based on forward and side scatter (FSC-A/SSC-A), followed by selection of single cells using FSC-A/FSC-H. The in vitro experiments included a viability stain (Zombie Violet Fixable Viability Kit, BioLegend), confirming that > 98% of gated lymphocytes were viable.

Spike protein expression levels on the cell membrane were quantified by flow cytometry at 24h and 48 hours after transfection into HEK293T cells. For flow cytometric analysis, cells were harvested and stained with a viability dye for 5 min. After washing, cells were incubated with anti-spike receptor-binding domain (RBD) chimeric monoclonal antibody (1:100 dilution, Sino Biological, 40150-D001) for 30 min, followed by washing and incubation with PE-anti-human IgG Fc (1:100 dilution, BioLegend, 410708) for 30 min. Samples were analyzed on BD FACSVerse.

In vitro cytotoxicity of CD19 CAR T-cells

NALM-6 (constitutively expressing the firefly luciferase) target cells were seeded into 96-well plates and co-cultured with CD19 CAR T cells at the indicated ratios of for up to 7 days. Cells were washed with DPBS, lysed and assayed for firefly luciferase activity according to the manufacturer's instructions (Promega, E2620).

LNP encapsulation of the RNAs

The mRNAs or circRNAs were encapsulated in lipid nanoparticles (LNPs). The RNAs were first diluted in citrate buffer (pH 4.0) to a final concentration of 90 μg/ml. The lipids (the molar ratio of lipids in this formulation is ALC-0315: DSPC: Cholesterol: ALC-0159 = 46.3: 9.4: 42.7: 1.6) were mixed with the RNA solution at the volume ratio of 1:3 using INano L+ (Micro&Nano). The LNP-RNA formulations were diluted 40-fold in 1×PBS buffer (pH 7.2-7.4) and concentrated by ultrafiltration with Amicon Ultra Centrifugal Filter Unit (Merck). The RNA concentration and encapsulation efficiency were measured using the Quant-it RiboGreen RNA Assay Kit (Invitrogen, R11490). The particle sizes were measured by dynamic light scattering using Zetasizer Lab (Malvern).

In vivo delivery of mRNA and circRNA

The mouse studies were conducted in strict accordance with the guidelines set by the Chinese Regulations of Laboratory Animals and Laboratory Animal-Requirements of Environment and Housing Facilities. The protocols (20240017MSE) were approved by the Institutional Animal Care and Use Committee (IACUC) of Beijing Morningrise Hi-Tech Biotechnology Co. Mice were randomly assigned to experimental or control groups. The investigator was not blinded to assignments.

For mouse vaccination, groups of 6- to 8-week-old female C57BL/6J mice (n = 5/group) were intramuscularly immunized with either 1 μg or 3 μg LNP-formulated mRNAs or LNPs alone as a placebo control, using a 1-ml sterile syringe. A booster dose was administered 2 weeks later to boost immune responses. The sera were collected to measure the SARS-CoV-2-specific IgG antibody levels by ELISA as described below. The mouse experiments were conducted in accordance with ethical regulations and approved by local ethical committees.

For in vivo EPO expression studies, groups (n = 4/group) of 6- to 8-week-old female BALB/c mice (n = 4/group) were intravenously injected via the tail vein with either 3 μg LNP-formulated RNAs or LNPs alone as a placebo control as indicated or a placebo (LNP only), using a 1-ml sterile syringe. The Sera were collected at indicated time points, and the EPO levels were measured by ELISA as previously described. The mouse experiments were conducted in accordance with ethical regulations and approved by local ethical committees.

IgG antibody ELISA

All immunized mouse serum samples were heat-inactivated at 56°C for 30 min prior to use. Serial 2-fold dilutions of heat-inactivated sera, starting at 1:100, were added to 96-well plates (100 µl/well) coated with recombinant SARS-CoV-2 S antigen (Novoprotein, DRA49) and incubated for 2 hours at room temperature. After three washes with the wash buffer, horseradish peroxidase HRP-conjugated goat anti-mouse IgG antibody (1:10000 dilution) was added to the plates and incubated for 30 min at room temperature. The plates were washed for four times with the wash buffer. The HRP substrates and stop solution were added, and the absorbance at 450 nm with a 630 nm reference was measured using a TECAN Spark microplate reader. The antibody specific for the RBD of SARS-CoV-2 spike protein (Sinobiological, 40592-MP01) was used as a positive control. Endpoint titers were calculated as the reciprocal serum dilution yielding a signal 2.1-fold above background, based on a four-parameter logistic curve fit (*54*).

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ACKNOWLEDGMENTS

The authors thank other current or former employees, contractors, or consultants of Raina Biosciences who provide advice or assistance on the studies described in the manuscript. The authors thank J. Zha and C. Lai for their valuable discussion. **Funding:** The work is funded by Raina Biosciences. **Author contributions:** Conceived idea and designed the study: H.Z., T.K.L., J.C.; Designed the algorithms: H.Z.; Implemented the algorithm and performance bioinformatic analysis: H.Z., Y.X., H.H.; Designed and performed in vitro and in vivo experiments and analyzed data: J.C., H.Z., H.L., Y.L., Y.Q., Z.H., X.H.; Performed DNA construction, mRNA and circular RNA production and analysis, luciferase assays: J.W., H.W., L.M.; Wrote the paper: H.Z., T.K.L., J.C.; All authors discussed the results and edited the manuscript. **Competing interests:** H.Z. and J.C. have filed patent applications on the work. T.K.L. is a co-founder of Senti Biosciences, Synlogic, Engine Biosciences, Tango Therapeutics, Corvium, BiomX, Eligo Biosciences, Bota.Bio, NE47Bio, Raina Bio. T.K.L. also holds financial interests in nest.bio, CognitoHealth, Quark Biosciences, Personal Genomics, Lexent Bio, Pulmobiotics, Provectus Algae, Invaio, NSG Biolabs, PairX, NSG Ventures, Integrated Biosciences, Flagship VL43, Resvita Bio, Absci. H.Z., H.L., H.H., Y.L., J.W., Y.Q., H.W., L.M., Z.X., X.H., J.C. are employees of Raina Biosciences. Y.X. is the intern of Raina Biosciences. Other authors declare no competing interests. **Data and materials availability:** The mRNA sequences used in the biological experiments were listed in the supplementary materials and data S1. The training datasets and in silico analysis data are described in the supplementary materials and available in data S2 to S4. The code and models are available on Zenodo (*55*). The most updated code can be found at Github (https://github.com/RainaBio/GEMORNA).

SUPPLEMENTARY MATERIALS

science.org/doi/10.1126/science.adr8470
Supplementary Text
Figs. S1 to S15
Tables S1 and S2
Reference (*56*)
MDAR Reproducibility Checklist
Data S1 to S4

Submitted 18 July 2024; resubmitted 4 March 2025
Accepted 6 August 2025
Published online 28 August 2025
10.1126/science.adr8470

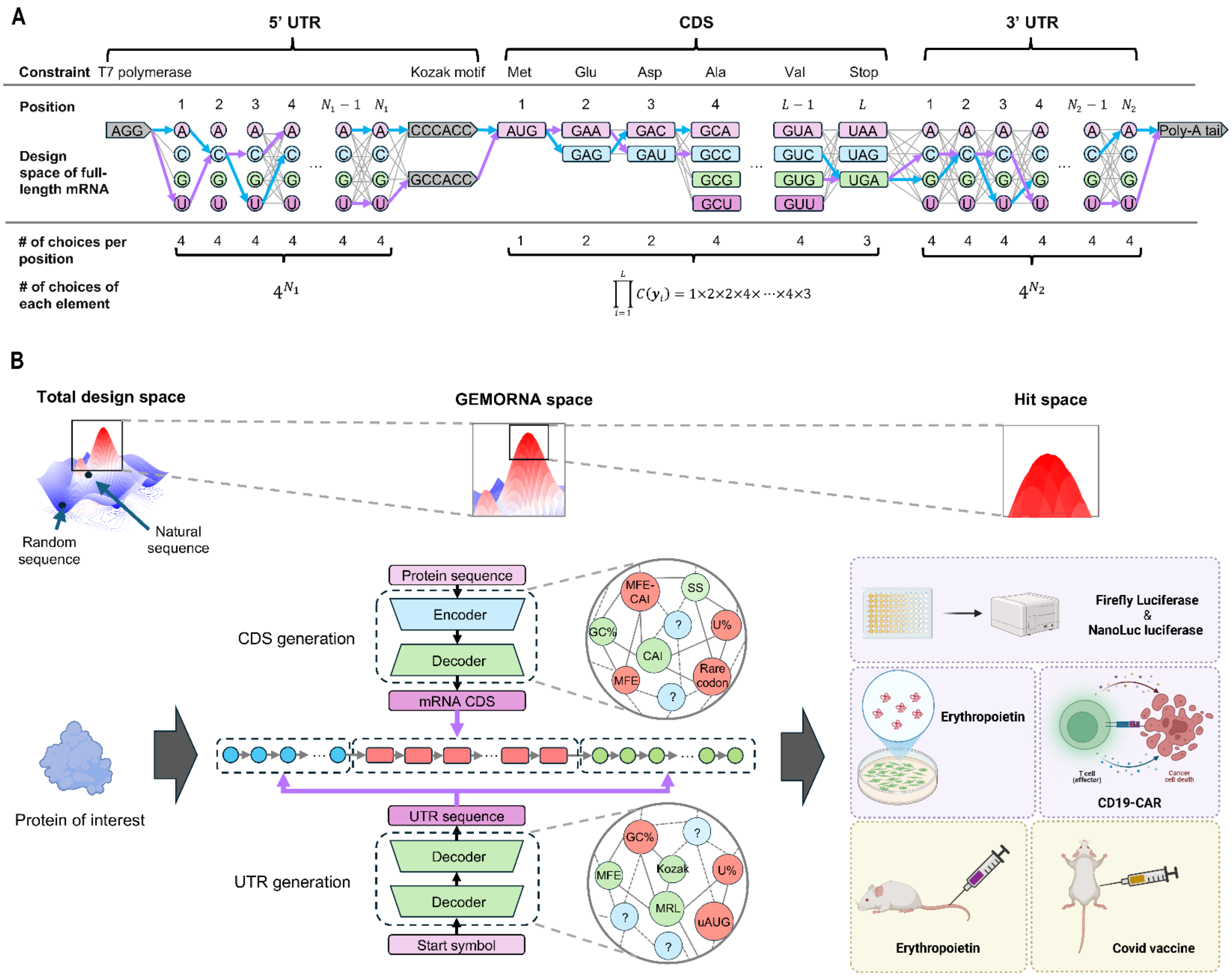
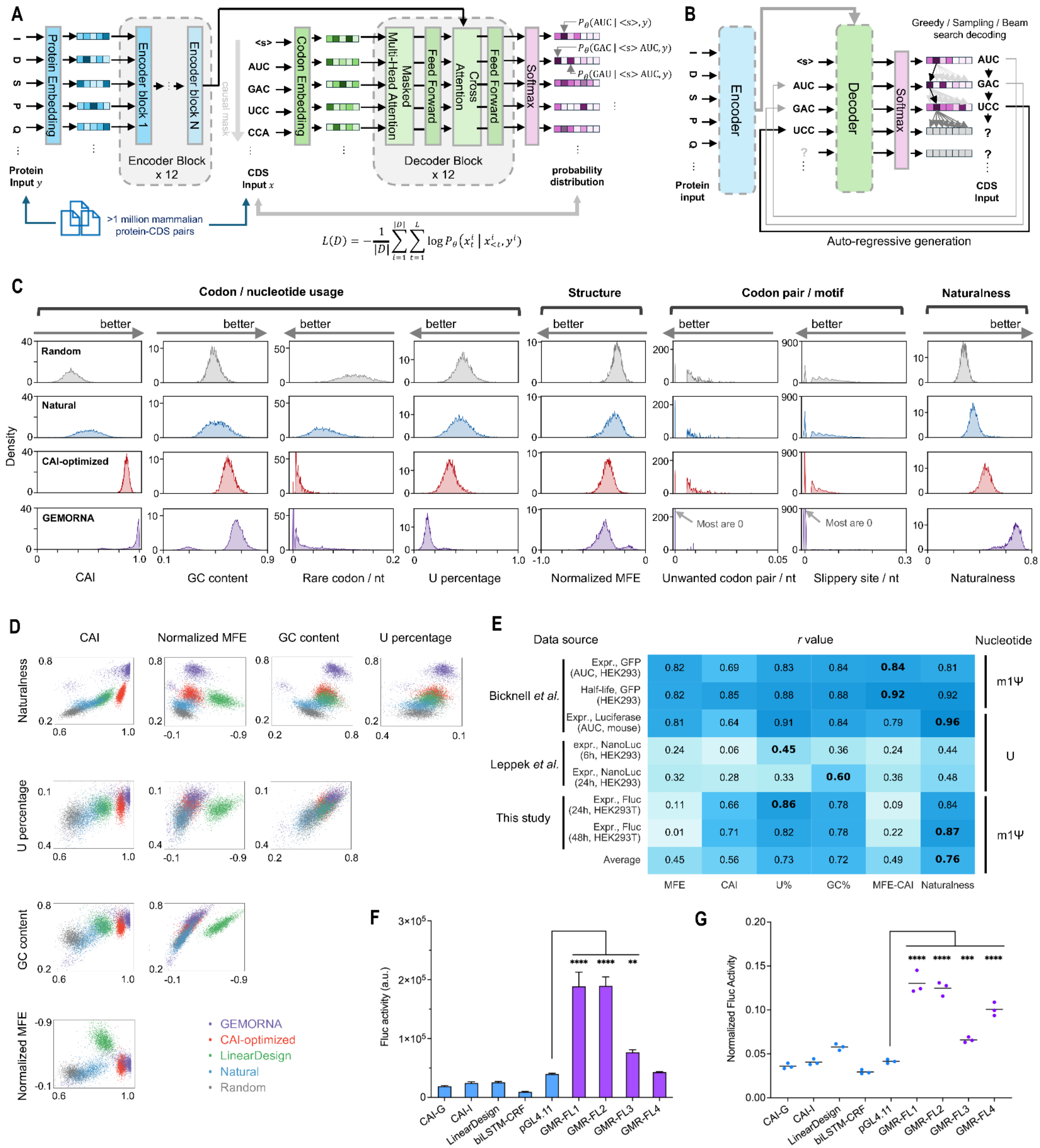


Fig. 1. Overview of designing full-length therapeutic mRNAs with GEMORNA. (**A**) Schematic representation of all possible mRNAs encoding firefly luciferase, including the 5′ UTR, CDS, and 3′ UTR sequences, illustrating the vast potential mRNA design space. Arrows between adjacent nucleotides (or codons) indicate possible choices at each position, with each path in the graph represents a unique full-length mRNA sequence with distinct UTR and CDS configurations. The highlighted blue path represents the natural alpha-globin UTRs with a codon-optimized CDS, while the purple path corresponds to GEMORNA's design. Note that the UTR's length (N_1 and N_2) can vary, whereas the CDS length (L) is constrained by the required protein length. $C(y_i)$ denotes the number of possible codon choice for a given amino acid y_i at each position. (**B**) GEMORNA-driven full-length mRNA design pipeline. The vast total design space is narrowed down by GEMORNA into a high-potency generation space, further validated through in vitro and in vivo experiments to reach hit space. To achieve this, GEMORNA models are trained to capture key features influencing mRNA expression and stability, ensuring that the GEMORNA mRNAs fall within the high-potency generation space.



A
Protein Embedding
Encoder block 1
Encoder block N
Encoder Block x 12
Protein Input y
causal mask
<s>
AUC
GAC
UCC
CCA
Codon Embedding
Masked Multi-Head Attention
Feed Forward
Cross Attention
Feed Forward
Softmax
Decoder Block x 12
CDS Input x
P_θ(AUC | <s>, y)
P_θ(GAC | <s> AUC, y)
P_θ(GAU | <s> AUC, y)
probability distribution
>1 million mammalian protein-CDS pairs
$L(D) = -\frac{1}{|D|}\sum_{i=1}^{|D|}\sum_{t=1}^{L}\log P_\theta(x_t^i \mid x_{<t}^i, y^i)$
B
Encoder
Decoder
Softmax
Greedy / Sampling / Beam search decoding
Protein input
CDS Input
Auto-regressive generation
C
Codon / nucleotide usage
Structure
Codon pair / motif
Naturalness
better
Random
Natural
CAI-optimized
GEMORNA
Density
Most are 0
CAI
GC content
Rare codon / nt
U percentage
Normalized MFE
Unwanted codon pair / nt
Slippery site / nt
Naturalness
D
CAI
Normalized MFE
GC content
U percentage
Naturalness
U percentage
GC content
Normalized MFE
GEMORNA
CAI-optimized
LinearDesign
Natural
Random
E
Data source
r value
Nucleotide
		MFE	CAI	U%	GC%	MFE-CAI	Naturalness	
Bicknell et al.	Expr., GFP (AUC, HEK293)	0.82	0.69	0.83	0.84	**0.84**	0.81	m1Ψ
	Half-life, GFP (HEK293)	0.82	0.85	0.88	0.88	**0.92**	0.92	
	Expr., Luciferase (AUC, mouse)	0.81	0.64	0.91	0.84	0.79	**0.96**	U
Leppek et al.	expr., NanoLuc (6h, HEK293)	0.24	0.06	**0.45**	0.36	0.24	0.44	
	Expr., NanoLuc (24h, HEK293)	0.32	0.28	0.33	**0.60**	0.36	0.48	
This study	Expr., Fluc (24h, HEK293T)	0.11	0.66	**0.86**	0.78	0.09	0.84	m1Ψ
	Expr., Fluc (48h, HEK293T)	0.01	0.71	0.82	0.78	0.22	**0.87**	
	Average	0.45	0.56	0.73	0.72	0.49	**0.76**	
F
Fluc activity (a.u.)
CAI-G
CAI-I
LinearDesign
biLSTM-CRF
pGL4.11
GMR-FL1
GMR-FL2
GMR-FL3
GMR-FL4
G
Normalized Fluc Activity

Fig. 2. The architecture of our GEMORNA-CDS model for CDS design, along with in silico analysis and in vitro experimental results of CDS sequences generated by GEMORNA-CDS. Compared to conventionally designed CDS sequences, GEMORNA-generated CDSs exhibited better codon usage and enhanced translational capacity. (**A** and **B**) Schematic visualization of the GEMORNA-CDS model for training (A) and inference (B). (**C** and **D**) One-dimensional (C) and two-dimensional (D) visualizations of various optimization objectives for CDSs from different sources. More than 3,700 proteins from were used as the target sequences. Random CDSs were generated by randomly selecting codons at each position. CAI-optimized CDSs were generated using high-frequency codons, ensuring a CAI > 0.9. (**E**) Correlations between the optimization objectives and experimental data on expression levels and half-life. (**F**) Expression levels of different Fluc2P CDSs at 48 hours in HEK293T cells (n = 3 biological replicates). Error bars represent the standard deviation of mean. GEMORNA-generated CDSs (abbreviated as "GMR-") were compared to a variety of CDSs generated from classical methods, while UTRs were kept unchanged. (**G**) Ratio of expression levels at 48 hours to expression levels at 24 hours in HEK293T cells as a measure of in-cell stability over time. One-way ANOVA was used for statistical analysis (**P < 0.01, ***P < 0.001, ****P < 0.0001).

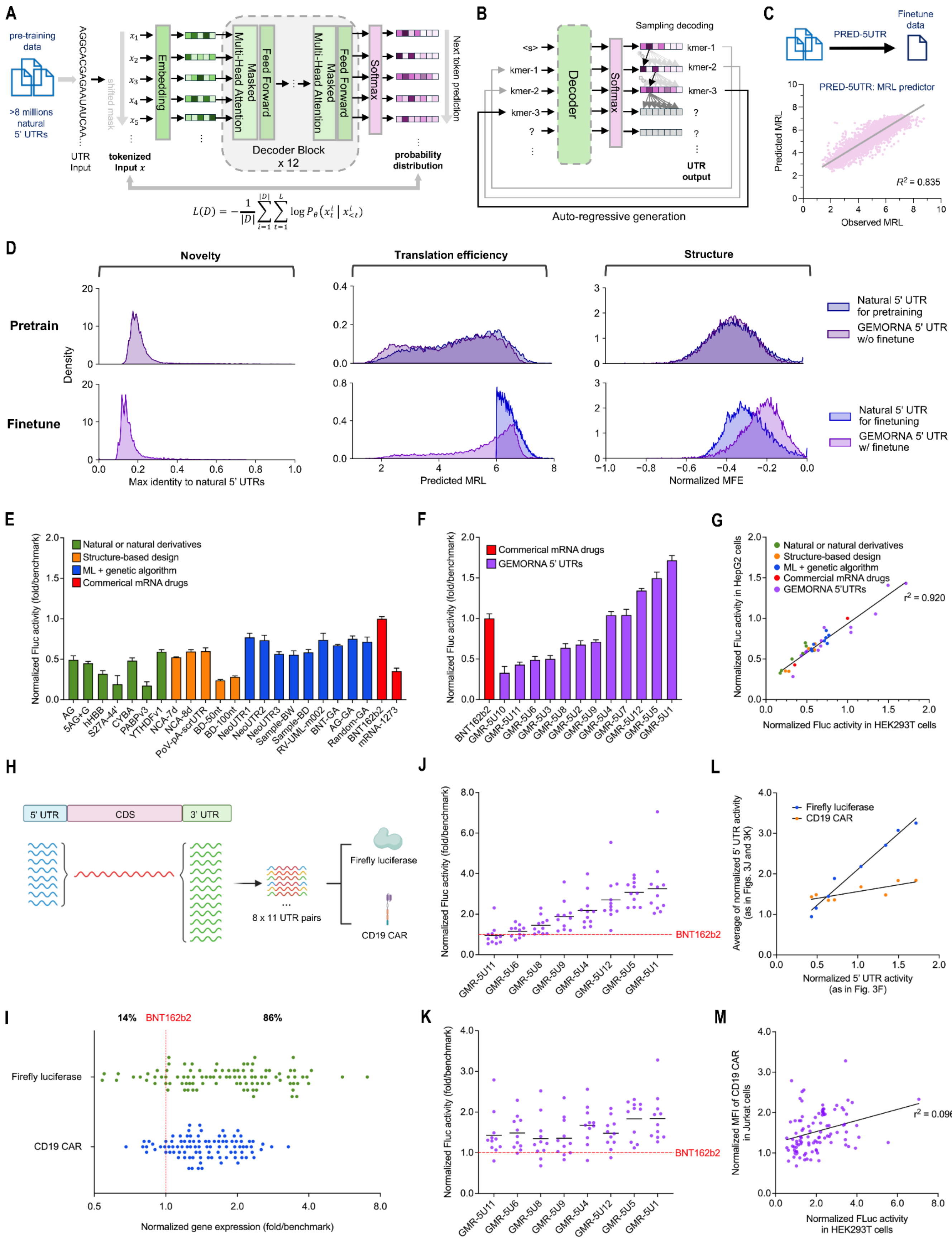

A
pre-training data
>8 millions natural 5' UTRs
AGGCACGAGAAUAUCAA...
UTR Input
shifted mask
tokenized input x
Embedding
Masked Multi-Head Attention
Feed Forward
Decoder Block x 12
Softmax
Next token prediction
probability distribution
$L(D) = -\frac{1}{|D|}\sum_{i=1}^{|D|}\sum_{t=1}^{L}\log P_\theta(x_t^i \mid x_{<t}^i)$
B
Sampling decoding
<s>
kmer-1
kmer-2
kmer-3
Decoder
Softmax
UTR output
Auto-regressive generation
C
PRED-5UTR
Finetune data
PRED-5UTR: MRL predictor
Predicted MRL
Observed MRL
R^2 = 0.835
D
Novelty
Translation efficiency
Structure
Pretrain
Finetune
Density
Max identity to natural 5' UTRs
Predicted MRL
Normalized MFE
Natural 5' UTR for pretraining
GEMORNA 5' UTR w/o finetune
Natural 5' UTR for finetuning
GEMORNA 5' UTR w/ finetune
E
Natural or natural derivatives
Structure-based design
ML + genetic algorithm
Commerical mRNA drugs
Normalized Fluc activity (fold/benchmark)
AG
5AG+G
hHBB
S27A-44'
CYBA
PABPv3
YTHDFv1
NCA-7d
NCA-8d
PoV-pA-scrUTR
BD-50nt
BD-100nt
NeoUTR1
NeoUTR2
NeoUTR3
Sample-BW
Sample-BD
RV-UML-m002
BNT-GA
AG-GA
Random-GA
BNT162b2
mRNA-1273
F
Commerical mRNA drugs
GEMORNA 5' UTRs
BNT162b2
GMR-5U10
GMR-5U11
GMR-5U6
GMR-5U3
GMR-5U8
GMR-5U2
GMR-5U9
GMR-5U4
GMR-5U7
GMR-5U12
GMR-5U5
GMR-5U1
G
Natural or natural derivatives
Structure-based design
ML + genetic algorithm
Commerical mRNA drugs
GEMORNA 5'UTRs
r^2 = 0.920
Normalized Fluc activity in HepG2 cells
Normalized Fluc activity in HEK293T cells
H
5' UTR
CDS
3' UTR
8 x 11 UTR pairs
Firefly luciferase
CD19 CAR
I
14%
BNT162b2
86%
Firefly luciferase
CD19 CAR
Normalized gene expression (fold/benchmark)
J
Normalized Fluc activity (fold/benchmark)
BNT162b2
GMR-5U11
GMR-5U6
GMR-5U8
GMR-5U9
GMR-5U4
GMR-5U12
GMR-5U5
GMR-5U1
K
Normalized Fluc activity (fold/benchmark)
BNT162b2
L
Firefly luciferase
CD19 CAR
Average of normalized 5' UTR activity (as in Figs. 3J and 3K)
Normalized 5' UTR activity (as in Fig. 3F)
M
Normalized MFI of CD19 CAR in Jurkat cells
Normalized FLuc activity in HEK293T cells
r^2 = 0.096

Fig. 3. The architecture of our GEMORNA-UTR model, and the in silico analysis and in vitro experimental results of UTR sequences generated by GEMORNA-UTR. (**A** and **B**) Schematic visualization of the GEMORNA-UTR model architecture at training time (A) and inference time (B). (**C**) Utilizing a predictive model, PRED-5UTR, to identify natural UTRs with high predicted mean ribosome load for fine-tuning. (**D**) Distributions of max identity, MRL and MFE between natural and GEMORNA 5′ UTRs. A small max identity indicates a low sequence similarity between our AI-generated UTRs and all natural UTRs in the reference database. (**E** and **F**) Relative Fluc activities of the benchmark 5′ UTRs (E) and the GEMORNA-generated 5′ UTRs (F), normalized to BNT162b2 5′ UTR. Error bars represent the standard deviation of mean. (**G**) Correlation of Fluc activities between HEK293T and HepG2 cells with varying 5′ UTRs. (**H**) Illustration of UTR pairs and their validation on Fluc2P and CD19-CAR targets. (**I**) Most UTR pairs exhibited enhanced expression levels than the BNT162b2 benchmark across both targets. (**J**) Normalized Fluc activities of UTR pair groups, where each 5′ UTR is paired with varying 3′ UTRs. (**K**) Normalized CD19 CAR expression levels of UTR pair groups. (**L**) 5′ UTR activities as shown in (F) compared to the mean of 5′ UTR activities presented in (J) and (K). (**M**) Correlation of UTR pair performance across two protein targets. The experiments were performed with $n = 3$ biological replicates.

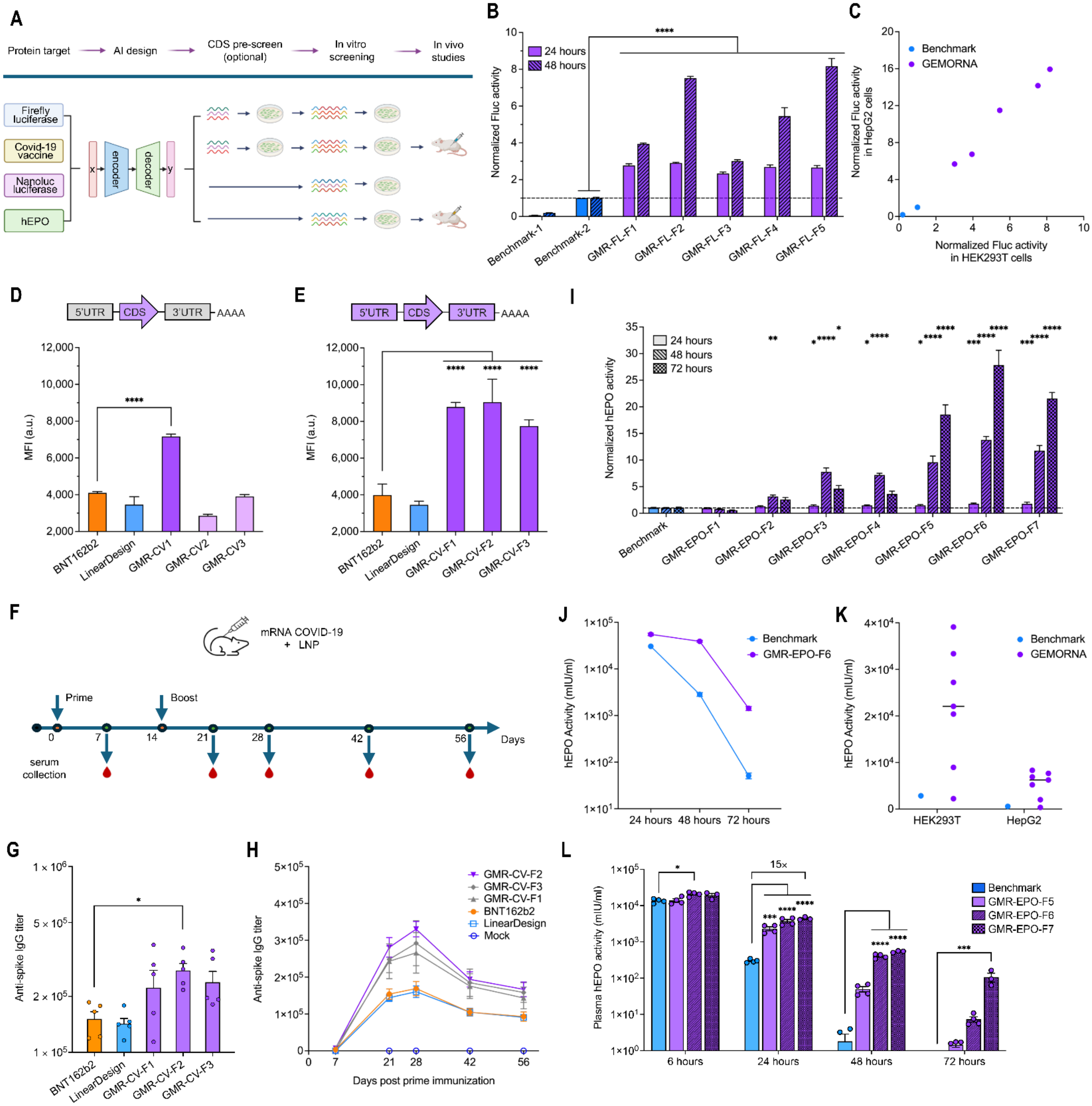
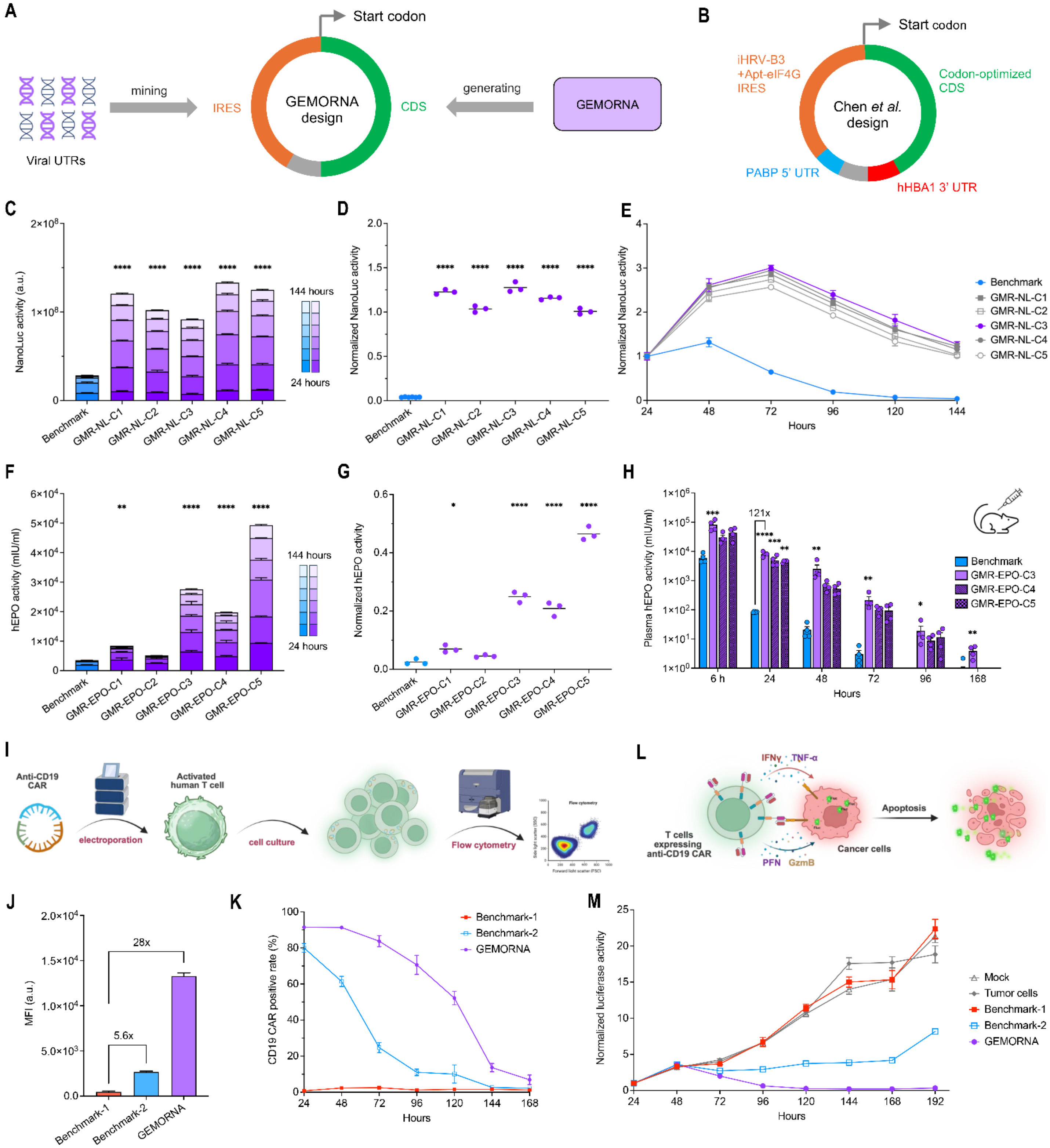

A
Protein target
AI design
CDS pre-screen (optional)
In vitro screening
In vivo studies
Firefly luciferase
Covid-19 vaccine
Nanoluc luciferase
hEPO
encoder
decoder
B
24 hours
48 hours
Normalized Fluc activity
Benchmark-1
Benchmark-2
GMR-FL-F1
GMR-FL-F2
GMR-FL-F3
GMR-FL-F4
GMR-FL-F5
C
Benchmark
GEMORNA
Normalized Fluc activity in HepG2 cells
Normalized Fluc activity in HEK293T cells
D
5'UTR
CDS
3'UTR
AAAA
MFI (a.u.)
BNT162b2
LinearDesign
GMR-CV1
GMR-CV2
GMR-CV3
E
GMR-CV-F1
GMR-CV-F2
GMR-CV-F3
I
24 hours
48 hours
72 hours
Normalized hEPO activity
Benchmark
GMR-EPO-F1
GMR-EPO-F2
GMR-EPO-F3
GMR-EPO-F4
GMR-EPO-F5
GMR-EPO-F6
GMR-EPO-F7
F
mRNA COVID-19 + LNP
Prime
Boost
Days
serum collection
J
hEPO Activity (mIU/ml)
Benchmark
GMR-EPO-F6
24 hours
48 hours
72 hours
K
HEK293T
HepG2
G
Anti-spike IgG titer
H
Mock
Days post prime immunization
L
Plasma hEPO activity (mIU/ml)
6 hours
24 hours
48 hours
72 hours
15×

Fig. 4. By combining GEMORNA-generated CDS and UTRs, we designed full-length mRNAs with stronger in vitro protein expression and in vivo potency compared with conventionally-designed counterparts and commercial products. (**A**) Summary of the targets and pipelines for full-length mRNA validation. (**B**) Normalized Fluc activities of seven full-length mRNAs encoding the Fluc2P reporter with varying UTRs and CDSs. (**C**) Normalized Fluc activities across two cell lines. (**D** to **H**) In vitro antigen expression and in vivo immunogenicity experimental results of COVID-19 mRNA vaccine. (D) In vitro expression level comparison of five optimized CDSs for the COVID-19 vaccine antigen. All sequences were constructed with BNT162b2's UTRs; LinearDesign represents the CDS of LinearDesign Seq-C (13), but with two changes: (1) two corresponding codons were altered to include the "2P" mutation in protein sequence, and (2) m1Ψ modification was used in place of unmodified uridine. (E) In vitro expression level comparison of full-length mRNA design for the COVID-19 vaccine antigen. (F) Timeline of injections and serum collection for in vivo COVID-19 mRNA experiment. (G) Comparison of anti-spike IgG titer on Day 21 at a 3 μg dose (n = 5 mice per group). (H) Time-course of anti-spike IgG titers through Day 56. (I-L) In vitro and in vivo protein expression of EPO mRNAs. (**I**) Normalized EPO activity comparison of eight full-length mRNAs in HEK293T cells. (**J**) Time-course of EPO activity through 72 hours post-transfection. (**K**) Distribution of EPO activity at 48 hours post-transfection in HEK293T and HepG2 cells, comparing GEMORNA designs to the benchmark. (**L**) Comparison of EPO expression in mice over time (n = 4 mice per group); values below 1 are not displayed. In vitro experiments were performed with n = 3 biological replicates. Error bars represent the standard deviation of mean for all in vitro data and in vivo EPO data. Error bars represent geometric error of geometric mean for in vivo COVID data. One-way ANOVA was used for statistical analysis (*$P < 0.05$, **$P < 0.01$, ***$P < 0.001$, ****$P < 0.0001$).



A
B
C
D
E
F
G
H
I
J
K
L
M
Start codon
mining
generating
Viral UTRs
IRES
CDS
GEMORNA design
GEMORNA
iHRV-B3 +Apt-eIF4G IRES
Codon-optimized CDS
Chen et al. design
PABP 5' UTR
hHBA1 3' UTR
NanoLuc activity (a.u.)
Normalized NanoLuc activity
144 hours
24 hours
Benchmark
GMR-NL-C1
GMR-NL-C2
GMR-NL-C3
GMR-NL-C4
GMR-NL-C5
Hours
hEPO activity (mIU/ml)
Normalized hEPO activity
Plasma hEPO activity (mIU/ml)
GMR-EPO-C1
GMR-EPO-C2
GMR-EPO-C3
GMR-EPO-C4
GMR-EPO-C5
121x
Anti-CD19 CAR
electroporation
Activated human T cell
cell culture
Flow cytometry
T cells expressing anti-CD19 CAR
IFNγ
TNF-α
PFN
GzmB
Cancer cells
Apoptosis
MFI (a.u.)
28x
5.6x
Benchmark-1
Benchmark-2
GEMORNA
CD19 CAR positive rate (%)
Normalized luciferase activity
Mock
Tumor cells

Fig. 5. The GEMORNA platform is able to generate circRNAs with improved expression level, durability and potency compared with conventionally-designed and patented counterparts. (**A**) Our GEMORNA circRNA topology integrates with mined IRESs and AI-generated CDSs. (**B**) The optimized circRNA topology proposed by Chen *et al*. which uses screened or optimized UTRs, IRES and CDSs; this topology was used as a benchmark compared with our GEMORNA-enabled design approach. (**C** to **E**) Comparison of NanoLuc luciferase expression levels and expression durability between GEMORNA circRNAs and a benchmark circRNA in HEK293T cells. Accumulated NanoLuc luciferase within 144 hours post-transfection (C), ratios of NanoLuc product at 144 hours to product at 24 hours (D), and normalized NanoLuc luciferase activities from 24 hours to 144 hours (E) are shown. (**F** to **H**) Comparison between GEMORNA circRNAs and benchmark circRNA expressing EPO protein in HEK293T cells and in mice. Accumulated EPO protein levels over 144 hours are shown in (F), and expression level ratios of 144 hours to 24 hours are also shown in (G). EPO protein expression levels in mice (n = 4 mice per group) are shown in (H); values below 1 are not displayed. (I-M) Expression levels, durability and anti-tumor efficacy of three circRNAs expressing CD19 CAR. (**I**) Schematic illustration of the in vitro assay used to measure CD19 CAR activity in T cells. (**J**) Mean Fluorescence Intensity (MFI) of three circRNAs at 24 hours post-transfection. (**K**) Time-course of CD19 CAR positive rate through 168 hours post-transfection. (**L**) Schematic illustration of the PBMC assay used to measure anti-tumor efficacy. (**M**) Comparisons of in vitro T cell-mediated killing of NALM6 cells (E:T=1:6) between GEMORNA circRNAs and two control mRNAs. "Tumor cells": no T cell-mediated treatment. "Mock": T cell-mediated treatment without CD19 CAR circRNAs transfected. "Benchmark-1", "Benchmark-2" and "GEMORNA": T cell-mediated treatments with transfected codon-optimized circRNAs, patented circRNAs and GEMORNA circRNAs, respectively; all these circRNAs encode CD19 CAR. In vitro experiments were performed with n = 3 biological replicates. Error bars represent the standard deviation of mean. One-way ANOVA was used for statistical analysis (*P < 0.05, **P < 0.01, ***P < 0.001, ****P < 0.0001).